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18/133,639	04/12/2023	Tetsuya SHIBATA	224232	8276

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OLIFF PLC
P.O. BOX 320850
ALEXANDRIA, VA 22320-4850

EXAMINER
ANDREWS, FELIX BRYAN

ART UNIT	PAPER NUMBER
2812	

NOTIFICATION DATE	DELIVERY MODE
08/25/2025	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

OfficeAction25944@oliff.com
jarmstrong@oliff.com

Office Action Summary	Application No. 18/133,639	Applicant(s) SHIBATA et al.	
	Examiner FELIX B ANDREWS	Art Unit 2812	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 12 April 2023.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-11 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-5 and 7-11 is/are rejected.

8) ☑ Claim(s) 6 is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☑ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 12 April 2023 is/are: a)☑ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☑ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☑ All b)☐ Some** c)☐ None of the:

1.☑ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED ACTION

Notice of Pre-AIA or AIA Status

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Priority

1. Receipt is acknowledged of certified copies of papers required by 37 CFR 1.55.

Specification

2. The title of the invention is not descriptive. A new title is required that is clearly indicative of the invention to which the claims are directed.

Claim Rejections - 35 USC § 103

3. In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis (i.e., changing from AIA to pre-AIA) for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

4. The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

5. The factual inquiries for establishing a background for determining obviousness under 35 U.S.C. 103 are summarized as follows:

1. Determining the scope and contents of the prior art.

2. Ascertaining the differences between the prior art and the claims at issue.

3. Resolving the level of ordinary skill in the pertinent art.

4. Considering objective evidence present in the application indicating obviousness or nonobviousness.

6. This application currently names joint inventors. In considering patentability of the claims the examiner presumes that the subject matter of the various claims was commonly owned as of the effective filing date of the claimed invention(s) absent any evidence to the contrary. Applicant is advised of the obligation under 37 CFR 1.56 to point out the inventor and effective filing dates of each claim that was not commonly owned as of the effective filing date of the later invention in order for the examiner to consider the applicability of 35 U.S.C. 102(b)(2)(C) for any potential 35 U.S.C. 102(a)(2) prior art against the later invention.

7. Claim 1 is rejected under 35 U.S.C. 103 as being unpatentable over Degawa et al. (US 2019/0305215) [Hereinafter Degawa] & Ichimura et al. (US 2004/0207961) [Hereinafter Ichimura].

Regarding claim 1, Degawa teaches A light detection element comprising:

A magnetic element **[**fig. 8, magnetoresistance element 101, para 8**]**,

a capacitor **[**fig. 8, para 121, *"a capacitor may be provided in the output signal line **122** (as an example, the output signal line **122** between a part of an inductor **125** connected to the output signal line **122** and the second port **121**) between a part constituting a closed circuit with a power supply **127**, the output signal line **122**, the magnetoresistance effect element **101**, a line **124**, and the ground G and the second port **121**. When a capacitor is provided in the part, it is possible to avoid applying an invariant component of a current to an output signal output from the second port **121**."***]***,**

and a resistor **[**fig. 8, para 143, *"a resistance element may be used in place of the inductor **125"***]**,**

wherein the magnetic element (fig. 1A/8, 101) and the capacitor are connected in series [fig. 8, para 121; wherein the capacitor may be provided in the output signal line 122 thereby one of ordinary skill would understand the capacitor to be in series with the magnetic element 101],

the resistor is connected to the magnetic element and the capacitor in parallel [fig. 8, para 143],

the magnetic element includes a first ferromagnetic layer [fig. 1A/8, first ferromagnetic layer 101A, para 8] ,

a second ferromagnetic layer [fig. 1A/8, second ferromagnetic layer 101B, para 8],

and a spacer layer [fig. 1A/8, spacer layer 101C, para 8] sandwiched between the first ferromagnetic layer and the second ferromagnetic layer [fig. 1/8].

Degawa fails to explicitly disclose the magnetic element is configured to be irradiated with a light containing an optical signal with a change of intensity of the light.

However Ichimura teaches wherein the magnetic element [fig. 4, MR device 40, para 49] is configured to be irradiated with a light containing an optical signal with a change of intensity of the light [para 90].

Therefore it would have been obvious to someone of ordinary skill in the art before the effective filing date of the claimed invention for the magnetic element to be configured to be irradiated with a light containing an optical signal with a change of intensity of the light to optically tune the magnetic properties leading to changes in electrical resistance.

8. Claims 2-3, & 5 are rejected under 35 U.S.C. 103 as being unpatentable over Degawa & Ichimura as applied to claim 1 and further in view of Zhu et al. (US 2019/0157345) [Hereinafter Zhu].

Regarding claim 2, Degawa/Ichimura teaches The light detection element according to claim 1,further comprising

a first electrode layer **[Degawa,** fig. 1A, electrode wiring 107, para 9**]** and

wherein the magnetic element **(Degawa,** fig. 1A/8, 101**)** has a first surface **(**top surface**)** and a second surface **(**bottom surface**)** opposing each other in a laminating direction of the magnetic element **(Degawa,** fig. 1A/8, 101**)**,

the first electrode layer **(Degawa,** fig. 1A/8, 107**)** is electrically connected to the first surface **(**top surface**)**.

Degawa/Ichimura fails to explicitly disclose a first dielectric body,

a slit filled with the first dielectric body is provided in the first electrode layer,

and the first dielectric body and the first electrode layer constitute at least a part of the capacitor.

However, Zhu teaches a first dielectric body **[**annotated fig. 23b, interlayer dielectric layer 1055, para 92**]**,

a slit filled with the first dielectric body **[annotated** fig. 23b, opening between layers 1053/1051/1049**]** is provided in the first electrode layer **[**fig. 23b, para 91; wherein *"an upper electrode layer may be formed over the free layer **1053"*** **]**,

and the first dielectric body **(**fig. 23b, 1055**)** and the first electrode layer **(**fig. 23b, para 91; wherein *"an upper electrode layer may be formed over the free layer **1053"**)* constitute at least a part of the capacitor **[**fig. 23b, wherein the MTJ structure 1055/1053/1051/1049/upper & lower electrodes constitute a capacitive structure**]**.

Therefore it would have been obvious to someone of ordinary skill in the art before the effective filing date of the claimed invention for the device to comprise a first dielectric body to provide electrical insulation preventing direct current flow ensuring reliability of the device.

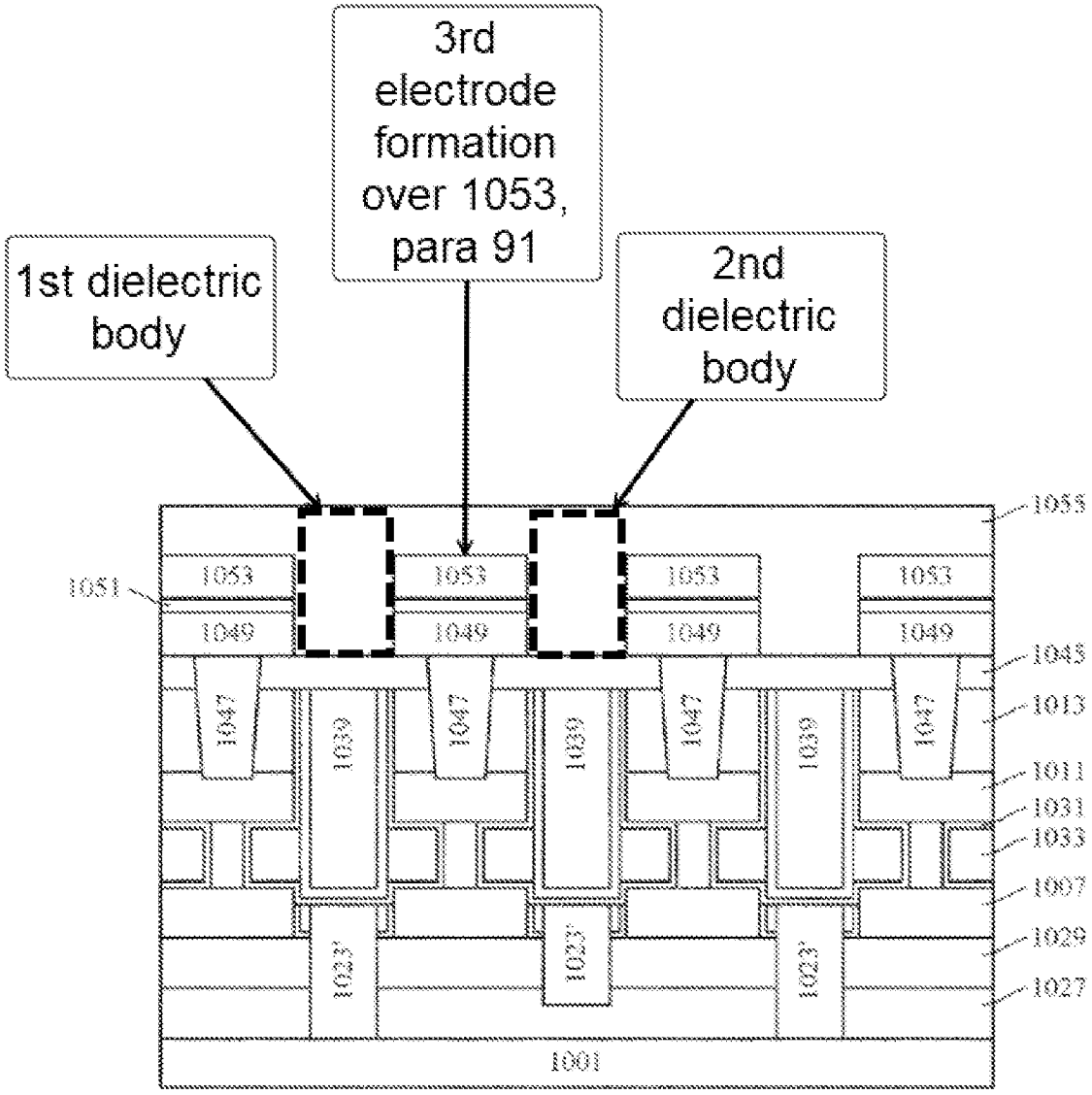


FIG. 23(b)

Zhu, Annotated Fig. 23b

Regarding claim 3, Degawa/Ichimura/Zhu teaches The light detection element according to claim 2, further comprising

a first connecting via **[Zhu,** fig. 23b, contacts 1047, para 89**]** and

a second electrode layer **[Degawa,** fig. 1A, electrode wiring 108, para 9**]**

wherein the resistor is connected to the first electrode layer branched off from the first electrode layer at a connecting part **[Degawa,** fig. 8, para 143, junction point of signal line 122 and resistive element 125**]**,

the resistor is located between the connecting part of the first electrode layer **[Degawa,** fig. 8, junction point of signal line 122 and resistive element 125**]** and the first connecting via **[**wherein one of ordinary skill in the art would understand the combination of Degawa and Zhu as utilized in claim 2 rejection would render the resistor to be between the connecting part and the first connecting via under the second electrode layer**]**,

the slit is located between the magnetic element and the connecting part **[**wherein one of ordinary skill in the art would understand the combination of Degawa and Zhu as utilized in claim 2 rejection would render the slit **(**openings between magnetic elements of Zhu fig. 23b**)** in

combination with Degawa device layout of fig. 8 renders the claimed limitation**]**, and

the second electrode layer **(Degawa,** fig. 8, 108**)** is electrically connected to the second surface **(**bottom surface**)** and the first connecting via **[**Zhu teaches para 91, *"a lower electrode layer may be formed below the fixed layer **1049"*** thereby the second electrode layer 1049 is connected to the bottom surface of the magnetic element and the first connecting via 1047**]**,

and electrically connected to the first electrode layer via the first connecting via and the resistor **[Degawa,** fig. 1/8; wherein the second electrode **(**108**)** is connected to the first electrode layer **(**107**)**.

Regarding claim 5, Degawa/Ichimura teaches The light detection element according to claim 1; further comprising

wherein the magnetic element **(**fig. 1/8, 101**)** has a first surface **(**top surface**)** and a second surface **(**bottom surface**)** opposing each other in a laminating direction of the magnetic element **(**fig. 1/8**)**.

Degawa/Ichimura fails to explicitly disclose a third electrode layer and

a second dielectric body,

a position of the second dielectric body is between the third electrode layer and the second surface in the laminating direction

and the second dielectric body and the third electrode layer constitute at least a part of the capacitor.

However, Zhu teaches a third electrode layer **[**fig. 23b, para 91; wherein *"an upper electrode layer may be formed over the free layer **1053"***]**, and**

a second dielectric body **[**annotated fig. 23b, interlayer dielectric layer 1055, para 92**]**,

a position of the second dielectric body is between the third electrode layer and the second surface **(**bottom surface**)** in the laminating direction **[**annotated fig. 23b**]**

and the second dielectric body and the third electrode layer constitute at least a part of the capacitor **[**fig. 23b, wherein the MTJ structure 1055/1053/1051/1049/upper & lower electrodes constitute a capacitive structure**]**.

Therefore it would have been obvious to someone of ordinary skill in the art before the effective filing date of the claimed invention for the device to comprise a second dielectric body to provide electrical insulation preventing direct current flow ensuring reliability of the device.

9. Claim 4 is rejected under 35 U.S.C. 103 as being unpatentable over Degawa, Ichimura, & Zhu as applied to claims 2-3 and further in view of Iwata et al. (US 2009/0219638) [Hereinafter Iwata].

Regarding claim 4, Degawa/Zhu teaches The light detection element according to claim 2,

Degawa/Zhu fails to explicitly disclose wherein the magnetic element is configured to be irradiated with the light from a side of the first surface.

However, Iwata teaches wherein the magnetic element **[**fig. 1, magnetic sensing section 7, para 42**]** is configured to be irradiated with the light **[**fig. 1, near field light 16, para 43**]** from a side of the first surface **(**top surface**) [**fig. 1**]**.

Therefore it would have been obvious to someone of ordinary skill in the art before the effective filing date of the claimed invention for the magnetic element to be irradiated with light on the top surface to detect the presence and intensity of light wherein magneto-optical effects can lead to enhanced sensitivity and resolution compared to traditional sensors.

10. Claim 7 is rejected under 35 U.S.C. 103 as being unpatentable over Degawa, Ichimura, Zhu as applied to claims 2-3, 5 and further in view of Iwata.

Regarding claim 7, Degawa/Ichimura/Zhu teaches The light detection element according to claim 5.

Degawa/Ichimura/Zhu fails to explicitly disclose wherein the magnetic element is, configured to be irradiated with the light from a side of the first surface.

However, Iwata teaches wherein the magnetic element **(**fig. 1, magnetic sensing section 7, para 42**]** is configured to be irradiated with the

light **[**fig. 1, near field light 16, para 43**]** from a side of the first surface **(**top surface**) [**fig. 1**]**.

Therefore it would have been obvious to someone of ordinary skill in the art before the effective filing date of the claimed invention for the magnetic element to be irradiated with light on the top surface to detect the presence and intensity of light wherein magneto-optical effects can lead to enhanced sensitivity and resolution compared to traditional sensors.

11. Claim 8 is rejected under 35 U.S.C. 103 as being unpatentable over Degawa, Ichimura, & Yamane et al. (US 2019/0245254) [Hereinafter Yamane].

Regarding claim 8, Degawa teaches A light detection element comprising:

a magnetic element **[**fig. 8, magnetoresistance element 101, para 8**]**,

a resistor **[**para 128, *"For the inductor **125**, a resistance element including a chip inductor…"***]**,

and an inductor **[**para 128, *"For the inductor **125**, a resistance element including a chip inductor…"***]**;

the magnetic element includes a first ferromagnetic layer **[**fig. 1A/8, first ferromagnetic layer 101A, para 8**]** ,

a second ferromagnetic layer **[**fig. 1A/8, second ferromagnetic layer 101B, para 8**]**,

and a spacer layer **[**fig. 1A/8, spacer layer 101C, para 8**]** sandwiched between the first ferromagnetic layer and the second ferromagnetic layer **[**fig. 1/8**]**,

Degawa fails to explicitly disclose the magnetic element is configured to be irradiated with a light containing an optical signal with a change of intensity of the light.

However Ichimura teaches wherein the magnetic element **[**fig. 4, MR device 40, para 49**]** is configured to be irradiated with a light containing an optical signal with a change of intensity of the light **[**para 90**]**.

Therefore it would have been obvious to someone of ordinary skill in the art before the effective filing date of the claimed invention for the magnetic element to be configured to be irradiated with a light

containing an optical signal with a change of intensity of the light to optically tune the magnetic properties leading to changes in electrical resistance.

Degawa/Ichimura fails to explicitly disclose wherein the magnetic element and the resistor are connected in series,

the inductor is connected to the magnetic element and the resistor in parallel,

However, Yamane teaches wherein the magnetic element **(**fig. 5, magnetoresistive elements 1a/1b, para 35**]** and the resistor **[**fig. 5, current source 13a, para 46; wherein the direct-current source may be composed of a circuit which is capable of generating constant DC current and in which a fixed/variable resistor is combined with a DC voltage source; which behaves as a resistor limiting the current flow**]** are connected in series,

the inductor **[**fig. 5, inductor 15, para 79**]** is connected to the magnetic element **(**fig. 5, 1a/1b**)** and the resistor **(**fig. 5, 13a; wherein the current source is not ideal, which limits current flow and behaves as a resistor**)** in parallel **(**fig. 5**)**.

Therefore it would have been obvious to someone of ordinary skill in the art before the effective filing date of the claimed invention for the device to comprise the circuit topology as taught by Yamane to help

stabilize the current protecting the device from damage due to excessive current while also acting as a filter blocking unwanted transients in the light detectors output signal creating a more stable and readable output.

12. Claims 9-10 are rejected under 35 U.S.C. 103 as being unpatentable over Degawa, Ichimura, & Kawabata et al. (US 2017/0238834) [Hereinafter Kawabata].

Regarding claim 9, Degawa teaches A receiving device comprising

a magnetic element **[**fig. 8, magnetoresistance element 101, para 8**]**,

and a circuit **[**fig. 8**]**,

wherein the magnetic element includes a first ferromagnetic layer **[**fig. 1A/8, first ferromagnetic layer 101A, para 8**]**,

a second ferromagnetic layer **[**fig. 1A/8, second ferromagnetic layer 101B, para 8**]**,

and a spacer layer **[**fig. 1A/8, spacer layer 101C, para 8**]** sandwiched between the first ferromagnetic layer and the second ferromagnetic layer **[**fig. 1/8**]**.

the circuit is connected to the magnetic element and is configured to output a voltage with a magnitude corresponding to an amount of change per unit time of an output voltage output from the magnetic element according to the optical signal **[**fig. 8, para 61, **[***"In addition, at least one of the first electrode wiring* ***107*** *and the second electrode wiring* ***108*** *transmits a signal output from the magnetoresistance effect element* ***101****. For example, a direct current or a DC voltage may be applied to the magnetoresistance effect element* ***101*** *through the first electrode wiring* ***107*** *and the second electrode wiring* ***108****. In addition, for example, the second electrode wiring* ***108*** *may transmit a signal (a high-frequency voltage or a high-frequency current) output from the magnetoresistance effect element* ***101****."***]**.

Degawa fails to explicitly disclose the magnetic element is configured to be irradiated with a light containing an optical signal with a change of intensity of the light.

However Ichimura teaches wherein the magnetic element **[**fig. 4, MR device 40, para 49**]** is configured to be irradiated with a light containing an optical signal with a change of intensity of the light **[**para 90**]**.

Therefore it would have been obvious to someone of ordinary skill in the art before the effective filing date of the claimed invention for the magnetic element to be configured to be irradiated with a light containing an optical signal with a change of intensity of the light to optically tune the magnetic properties leading to changes in electrical resistance.

Degawa/Ichimura fails to explicitly disclose the receiving device is configured to receive the optical signal on a basis of the voltage output from the circuit.

While Degawa notes the optical signal to be a high-frequency voltage or current output that is sent to an output terminal to be transmitted to the outside.

Kawabata teaches a receiving device is configured to receive the optical signal on a basis of the voltage output from the circuit **[**para 144**]**.

Therefore it would have been obvious to someone of ordinary skill in the art before the effective filing date of the claimed invention for the device to comprise a receiving device for the optical signal from the circuit to convert optical signals into electrical signals enabling extraction, amplification, and processing of the information carried by the light signals.

Regarding claim 10, Degawa/Ichimura/Kawabata teaches The receiving device according to claim 9,

wherein the circuit includes a capacitor connected to the magnetic element in series **[Degawa,** fig. 8, para 121**]**,

and a resistor connected to the magnetic element and the capacitor in parallel **[Degawa,** fig. 8, element 125, para 128**]**.

13. Claims 11 is rejected under 35 U.S.C. 103 as being unpatentable over Degawa, Ichimura, & Kawabata as applied to claims 9-10 and further in view of Yamane.

Regarding claim 11, Degawa/Ichimura/Kabata teaches The receiving device according to claim 9.

Degawa/Ichimura/Kabata fails to explicitly disclose wherein the circuit includes a resistor connected to the magnetic element in series,

and an inductor connected to the magnetic element and the resistor-in parallel.

However, Yamane teaches wherein the magnetic element **[**fig. 5, magnetoresistive elements 1a/1b, para 35**]** and the resistor **[**fig. 5, current source 13a, para 46; wherein the direct-current source may be composed of a circuit which is capable of generating constant DC current and in which a fixed/variable resistor is combined with a DC voltage source; which behaves as a resistor limiting the current flow**]** are connected in series,

the inductor **[**fig. 5, inductor 15, para 79**]** is connected to the magnetic element **(**fig. 5, 1a/1b**)** and the resistor **(**fig. 5, 13a; wherein the current source is not ideal, which limits current flow and behaves as a resistor**)** in parallel **(**fig. 5**)**.

Therefore it would have been obvious to someone of ordinary skill in the art before the effective filing date of the claimed invention for the device to comprise the circuit topology as taught by Yamane to help stabilize the current protecting the device from damage due to excessive current while also acting as a filter blocking unwanted transients in the light detectors output signal creating a more stable and readable output.

Allowable Subject Matter

14. Claim 6 is objected to as being dependent upon a rejected base claim, but would be allowable if rewritten in independent form including all of the limitations of the base claim and any intervening claims.

15. The following is a statement of reasons for the indication of allowable subject matter:

Regarding claim 6, While Degawa notes in para 138, *"While a case in which there is one magnetoresistance effect element **101** is an exemplary example as shown in FIG. 8, a plurality of magnetoresistance effect elements **101** may be used. In this case, the plurality of magnetoresistance effect elements **101** may be connected in parallel or may be connected in series. For example, when a plurality of magnetoresistance effect elements **101** having different ferromagnetic resonance frequencies are used, it is possible to widen a selection frequency band (pass frequency band). In addition, a configuration in which a high-frequency magnetic field generated in the output signal line **122** that outputs an output signal from one magnetoresistance effect element **101** is applied to another*

*magnetoresistance effect element **101** may be used. In such a configuration, since the output signal is filtered a plurality of times, it is possible to increase the filtering accuracy of the high-frequency signal."*

The prior art made of record fails to explicitly disclose The light detection element according to claim 5, further comprising

a second connecting via and a fourth electrode layer,

wherein the resistor is connected to the third electrode layer branched off from the third electrode layer at a connecting part

and the resistor is located between the connecting part and the second connecting, via,

and the fourth electrode layer is electrically connected to the first surface and the second connecting via

and electrically connected to the third electrode layer via the second connecting via and the resistor.

Thereby claim 6 contains allowable subject matter and would be allowable if rewritten in independent form including all of the limitations of the base claim and any intervening claims.

Conclusion

16. Any inquiry concerning this communication or earlier communications from the examiner should be directed to FELIX B ANDREWS whose telephone number is (703)756-1074. The examiner can normally be reached Monday - Friday 8:00 am - 5:00 pm ET.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, William Partridge can be reached at 571-270-1402. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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/FELIX B ANDREWS/
Examiner, Art Unit 2812

/William B Partridge/
Supervisory Patent Examiner, Art Unit 2812